

Olympiad Test Paper — Mathematics (Class 7) — Paper 5

Chapter 6: Number Play

Paper 5 — (progressively harder)

Subject	Mathematics (NCERT Class 7)
Chapter	6 — Number Play
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. An odd count of odd numbers forces an odd sum and parity survives every reversal; the Virahanka-Fibonacci rule adds the two previous terms (its parity repeats O,O,E with period 3 and its units digits with period 60); in the taller-in-front game a call at position k is at most $k-1$ and the sum of all calls counts taller-ahead pairs; a supercell strictly exceeds every edge-neighbour; Kaprekar fixes 6174 (4-digit) and 495 (3-digit); Collatz always falls to 1; digit-sum governs divisibility by 9 and 3 while the alternating digit-sum governs 11. Do not write on this paper — mark answers on your answer sheet.

1. A whole number N has digit-sum 23. What is the remainder when N is divided by 9, and what is the remainder when N is divided by 3?

- (A) remainder 2 on division by 9, remainder 2 on division by 3
- (B) remainder 5 on division by 9, remainder 0 on division by 3
- (C) remainder 5 on division by 9, remainder 2 on division by 3
- (D) remainder 0 on division by 9, remainder 2 on division by 3

2. The five-digit number $8A35B$ is divisible by 9, and its digit B is exactly 2 more than its digit A . What is the value of A ?

- (A) 3
- (B) 1
- (C) 5
- (D) 0

3. A number N leaves remainder 7 when divided by 11. The number M is formed by reversing the digits of the four-digit number N . Which statement is always true about M 's remainder when divided by 11?

- (A) M leaves remainder 4 (because reversing a 4-digit number negates its alternating digit-sum mod 11)
- (B) M also leaves remainder 7, since reversal never changes the value mod 11
- (C) M leaves remainder 0, since any number and its reverse differ by a multiple of 11
- (D) it cannot be determined without the actual digits

4. Ahaan multiplies three whole numbers and gets an EVEN product, then adds the same three numbers and gets an ODD sum. What is the only possible parity pattern of the three numbers?

- (A) exactly one of them is even (and two are odd)
- (B) all three are even
- (C) exactly two of them are even (and one is odd)
- (D) all three are odd

5. The numbers from 1 to 60 are written. How many of them have a digit-sum that is a multiple of 3?

- (A) 18
- (B) 21
- (C) 24
- (D) 20

6. A 3-digit number and its digit-reversal are both genuine 3-digit numbers (no leading zero on either). Their difference (larger minus smaller) is 297. What is the absolute difference between the hundreds digit and the units digit of the original number?

- (A) 2
- (B) 9
- (C) 3
- (D) 33

7. Kishor lists seven whole numbers whose product is odd and whose sum is then computed. What is the parity of that sum, and can the sum ever be even?

- (A) the sum is always even
- (B) the sum can be either, depending on the values
- (C) the sum is odd only if all seven equal 1
- (D) the sum is always odd and can never be even

8. A four-digit number 5 7 b c (first two digits 5 and 7) is divisible by both 9 and 11. What is the four-digit number?

- (A) 5715
- (B) 5742
- (C) 5760
- (D) 5709

9. How many numbers from 100 to 200 inclusive are divisible by 9?

- (A) 11
- (B) 12
- (C) 10
- (D) 9

10. A two-digit number equals 6 times the sum of its digits. The number is also even. What is the number?

- (A) 24 (B) 42
(C) 54 (D) 18

11. N is a six-digit number all of whose digits are the same nonzero digit d (a repdigit like 444444). For which value of d is N divisible by 7?

- (A) only $d = 7$ (B) only even digits d
(C) any nonzero digit d works (every six-digit repdigit is divisible by 7)
(D) no nonzero d makes it divisible by 7

12. A Virahanka-Fibonacci-type sequence (each term is the sum of the two before it) has 3rd term 11 and 6th term 47. What is its 1st term?

- (A) 3 (B) 4
(C) 5 (D) 2

13. In the Virahanka-Fibonacci sequence 1,1,2,3,5,8,13,21,34,... the units digits repeat with period 60. The units digit of the 7th term is 3. What is the units digit of the 67th term?

- (A) 8 (B) 3
(C) 1 (D) 6

14. A Virahanka-Fibonacci-type sequence has parity pattern O,O,E repeating with period 3 (starting 1,1,2,...). Among the first 100 terms, how many are EVEN?

- (A) 33 (B) 34
(C) 50 (D) 33 or 34 depending on the start

15. Two consecutive terms of a Virahanka-Fibonacci-type sequence are 72 and 117. What is the term FOUR places BEFORE the 72?

- (A) 18 (B) 9
(C) 27 (D) 12

16. Using the identity (sum of the first n Virahanka-Fibonacci terms) = (the (n+2)th term) - 1, what is $1+1+2+3+5+8+13+21+34+55+89+144$?

- (A) 376 (B) 375
(C) 232 (D) 377

17. A Virahanka-Fibonacci-type sequence starts with two positive whole numbers a and b with $a \leq b$, and its 5th term equals 30. How many different starting pairs (a, b) are possible?

- (A) 3 (B) 2
(C) 4 (D) 1

18. The number of ways to tile a 1-by-n strip with 1-tiles and 2-tiles is a Virahanka-Fibonacci value. A 1-by-7 strip has 21 tilings and a 1-by-8 strip has 34 tilings. How many tilings of a 1-by-10 strip use AT LEAST one 2-tile?

- (A) 89 (B) 55
(C) 87 (D) 88

- 19.** Each consecutive ratio of Virahanka-Fibonacci terms is about 1.618. Approximately how many times larger is the 15th term than the 9th term?
- (A) about 6 times (B) about 1.6 times
(C) about 1.618 to the 5th power, near 11 times (D) about 18 times
- 20.** A Virahanka-Fibonacci-type sequence begins with $a = 3$, $b = 3$, each later term the sum of the two before. Which is the FIRST term that is NOT a multiple of 3?
- (A) the 4th term (B) the 6th term
(C) there is no such term — every term is a multiple of 3
(D) the 7th term
- 21.** In the standard Virahanka-Fibonacci sequence 1,1,2,3,5,8,13,21,34,55,..., what is the parity of the SUM of the 1st through 17th terms?
- (A) odd (B) it depends on rounding
(C) even (D) cannot be determined
- 22.** Two consecutive Virahanka-Fibonacci-type terms are 105 and 170. What is the term immediately AFTER 170, and is it even or odd?
- (A) 275, which is odd (B) 275, which is even
(C) 265, which is odd (D) 65, which is odd
- 23.** In a taller-in-front line of 6 children with all distinct heights, the calls (front to back) are 0, 1, 2, 3, 4, 5. Treating each call as the count of taller children ahead, what is the total number of pairs (i, j) with i ahead of j and child i taller than child j ?
- (A) 21 (B) 15
(C) 5 (D) 10
- 24.** A taller-in-front line of 5 children of distinct heights produces calls (front to back) 0, 1, 2, 3, 4. Where does the SHORTEST child stand?
- (A) at the back (5th position) (B) at the front (1st position)
(C) in the middle (3rd position) (D) it cannot be determined
- 25.** In a taller-in-front line, the back child (position k in a line of k) calls a number equal to $k - 1$. What does this force about that back child?
- (A) the back child is the tallest in the line (B) the back child is exactly the median height
(C) the back child is the shortest in the entire line
(D) nothing certain can be said
- 26.** A taller-in-front line of 7 distinct-height children gives calls (front to back) 0, 0, 0, 0, 0, 0, 0. What is the total number of taller-ahead pairs?
- (A) 0 (B) 7
(C) 21 (D) 6
- 27.** A taller-in-front line of distinct heights has calls (front to back) 0, 1, 1, 2, 0. Exactly one of these calls is impossible because a call at position k can be at most $k - 1$. Which call is

impossible?

- (A) the 2nd child, who said 1
- (B) the 4th child, who said 2
- (C) none — every call obeys its position bound, so the sequence is possible
- (D) the 5th child, who said 0

28. Seven children of distinct heights line up. The taller-in-front calls are 0, 1, 2, 0, 1, 2, 3 (front to back). What is the total number of taller-ahead pairs (the inversion count)?

- (A) 9
- (B) 6
- (C) 12
- (D) 21

29. In a taller-in-front line of n distinct-height children, the calls are read and they happen to be 0, 1, 2, ..., $(n-1)$. For this exact pattern, how many children are taller than the MIDDLE-position child when $n = 9$?

- (A) 5
- (B) 8
- (C) 4
- (D) 0

30. A taller-in-front line of 4 distinct-height children has calls 0, 1, 0, 2. Reconstruct the heights and state the position of the TALLEST child.

- (A) the 1st position
- (B) the 4th position
- (C) the 3rd position
- (D) the 2nd position

31. In any taller-in-front line of n distinct-height children, what is the LARGEST possible value of the SUM of all the calls, and what arrangement achieves it?

- (A) $n(n+1)/2$, achieved when heights strictly increase
- (B) $n(n-1)/2$, achieved when heights strictly decrease from front to back
- (C) $n - 1$, achieved only by the back child calling out
- (D) n squared, achieved by any random arrangement

32. A taller-in-front line of 6 distinct-height children has a TOTAL call-sum of 14. Is such a configuration possible, given the maximum and minimum possible totals?

- (A) no — the total must always be a multiple of 6
- (B) no — 14 exceeds the maximum possible total of 10
- (C) yes, but only if two children share a height
- (D) yes — 14 lies between the minimum 0 and the maximum 15, and every integer in that range is achievable

33. A 3×3 grid is filled with 1-9 (each once). A supercell is strictly greater than every edge-neighbour. The cell holding 1 is placed to have the FEWEST neighbours possible. How many neighbours does it have, and can it ever be a supercell there?

- (A) 4 neighbours (the centre), and it can be a supercell
- (B) 3 neighbours (an edge-middle), and it cannot be a supercell

(C) 2 neighbours (a corner), and it can never be a supercell because 1 is the smallest value

(D) 2 neighbours (a corner), and it can be a supercell since corners are favoured

34. In a 3x3 magic square of 1-9 with magic sum 15, the four EDGE-MIDDLE cells (top, bottom, left, right) are known to be the four EVEN numbers 2, 4, 6, 8. What number sits in the centre?

(A) 1

(B) 9

(C) it varies with the arrangement

(D) 5

35. A 3x3 grid of 1-9 (no repeats) has its top row summing to 7 and its bottom row summing to 23. The middle row must therefore sum to which value, and is that even possible?

(A) 15, and it is possible since all three rows together use 1-9 summing to 45

(B) 14, and it is possible

(C) 15, but it is impossible because no three of 1-9 sum to 15 here

(D) 16, and it is possible

36. A 3x3 grid of 1-9 (no repeats) is arranged to MAXIMISE the number of supercells. The known maximum is 4. In such an optimal arrangement, where do the four supercells sit?

(A) at the four edge-middles

(B) at the four corners (each corner has only 2 neighbours, easiest to dominate)

(C) at the centre plus three corners

(D) all four in one row

37. A 3x3 magic square of 1-9 (magic sum 15) is first rotated 90 degrees and then reflected left-to-right. Is the final result still a magic square?

(A) no — combining two transformations destroys the diagonals

(B) no — reflection after rotation moves the centre off-centre

(C) only if the magic sum is odd

(D) yes — both rotation and reflection map rows, columns and diagonals onto rows, columns and diagonals, each still summing to 15

38. In a 3x3 grid of 1-9 (no repeats), the centre holds 5 and the two main diagonals sum to 15 and 19. What is the sum of the four corner cells?

(A) 20

(B) 24

(C) 26

(D) 21

39. A 3x3 grid of 1-9 (no repeats) has the four corner cells summing to 28. What is the largest the corners could possibly sum to, and is 28 achievable?

(A) the maximum is 28, so 28 is the only achievable value

(B) the maximum is 24, so 28 is impossible

(C) the maximum is 26, so 28 is impossible

(D) the maximum corner sum is 30, so 28 is achievable

40. A 3x3 grid of 1-9 (no repeats) is arranged to MINIMISE the value of its largest row sum. What is the smallest possible value of that largest row sum?

- (A) 14 (B) 15
(C) 16 (D) 24

41. A 3x3 grid of 1-9 (no repeats) has its top-left corner = 4, top row sum = 15, left column sum = 18, and centre = 5. The top row and left column share only the corner. What is the sum of the remaining two cells of the LEFT column (the centre 5 and the bottom-left)?

- (A) 11 (B) 14
(C) 13 (D) 9

42. A 3x3 grid of 1-9 (no repeats) is filled so that exactly the cells holding 9 and 8 are supercells. Which placement of 9 and 8 guarantees BOTH are supercells?

- (A) place 9 and 8 in non-adjacent cells (not sharing an edge), e.g. two opposite corners
(B) place 9 in the centre and 8 in a corner
(C) place 9 and 8 side by side in the same row
(D) place 8 in the centre and 9 in a corner

43. In a 3x3 magic square of 1-9 (magic sum 15), every cell value is replaced by (16 minus that value). Is the result still a magic square, and what is its magic sum?

- (A) yes, with magic sum 15 (unchanged) (B) no, the diagonals break
(C) yes, with magic sum 48 (D) yes, it is a magic square with magic sum 33

44. Run the Collatz process (halve if even, else $3n+1$) starting from 7. How many steps does it take to first reach 1?

- (A) 17 (B) 11
(C) 15 (D) 16

45. In the Collatz process started from 27, the sequence rises far before falling. What is the largest value it ever reaches before returning to 1?

- (A) 4616 (B) 6174
(C) 3077 (D) 9232

46. Kaprekar's 4-digit routine (descending minus ascending, repeat) is applied to 3524. After exactly TWO subtraction steps, what number appears?

- (A) 8352 (B) 3087
(C) 6174 (D) 2475

47. A number-machine replaces a number by the SUM of the SQUARES of its digits, repeating. Starting from 23, after how many steps does it first reach 1?

- (A) 5 steps (B) 3 steps
(C) it never reaches 1; it falls into an endless cycle
(D) 8 steps

48. A number-machine rule: if the number is even, halve it; if odd, subtract 3. Starting from 50, what value does the machine reach after exactly 6 steps?

- (A) 1 (B) 4
(C) 2 (D) 0

49. Run the Collatz process from 12. How many steps to reach 1, and how many of those steps are 'halving' (even) steps?

- (A) 9 steps, of which 7 are halving steps (B) 9 steps, of which 9 are halving steps
(C) 10 steps, of which 7 are halving steps (D) 8 steps, of which 6 are halving steps

50. Kaprekar's 3-digit routine (descending minus ascending, repeat) fixes 495. Starting from 297, how many subtraction steps reach 495?

- (A) 2 (B) 4
(C) 1 (D) 3

51. A 'digit-sum machine' replaces a number by its digit-sum repeatedly until a single digit remains (the digital root). What is the digital root of the product $4 \times 7 \times 11 \times 13$ (which equals 4004)?

- (A) 9 (B) 2
(C) 4 (D) 8

52. How many 4-digit palindromes (form abba) are divisible by 11?

- (A) 90 (B) 9
(C) 18 (D) 45

53. What is the largest 4-digit palindrome (form abba) that is divisible by both 4 and 9?

- (A) 9999 (B) 8888
(C) 6336 (D) 9009

54. A two-digit number added to its digit-reversal gives 121, and its two digits differ by 3. What is the larger of the original number and its reversal?

- (A) 74 (B) 85
(C) 63 (D) no such two-digit number exists

55. Using the digits 2, 4, 6, 8 each exactly once to form a 4-digit number, what is the difference between the largest and smallest such numbers, and is that difference itself divisible by 9?

- (A) 6174, and yes it is divisible by 9 (B) 6084, and yes it is divisible by 9
(C) 6174, but it is not divisible by 9 (D) 5994, and yes it is divisible by 9

56. A 3-digit palindrome aba is divisible by 4. Given that a palindrome aba ends in the digit a, for how many such palindromes (a from 1-9, b from 0-9) is the number divisible by 4?

- (A) 18 (B) 10
(C) 20 (D) 25

57. Take any 3-digit number with first digit strictly greater than its last digit. Reverse it and subtract the reverse from the original. Then reverse THAT result and add it to the result. What number do you always get?

- (A) 99
- (B) 198
- (C) 1000
- (D) 1089

58. How many palindrome numbers lie strictly between 1000 and 2000 (4-digit palindromes in that range)?

- (A) 9
- (B) 10
- (C) 11
- (D) 100

59. A 5-digit palindrome has the form $abcba$. How many such palindromes are divisible by 5?

- (A) 200
- (B) 100
- (C) 180
- (D) 90

60. Using each of the digits 1, 3, 5, 7, 9 exactly once to make a 5-digit number, how many of the arrangements give an ODD number, and what does that reveal?

- (A) 60 of them are odd
- (B) all 120 arrangements are odd, because every available digit is odd so the units digit is always odd
- (C) only 24 are odd
- (D) none are guaranteed odd

Solutions — Mathematics (Class 7), Chapter 6:

Number Play — Paper 5

Paper 5 — (progressively harder)

Marking: +4 correct, −1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	C	21	C	31	B	41	B	51	D
2	D	12	B	22	A	32	D	42	A	52	A
3	A	13	B	23	B	33	C	43	D	53	C
4	C	14	A	24	A	34	D	44	D	54	A
5	D	15	B	25	C	35	A	45	D	55	A
6	C	16	A	26	A	36	B	46	A	56	C
7	D	17	B	27	C	37	D	47	B	57	D
8	B	18	D	28	A	38	B	48	C	58	B
9	A	19	D	29	C	39	D	49	A	59	B
10	C	20	C	30	C	40	B	50	D	60	B

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (C) A number is congruent to its digit-sum for both 9 and 3. 23 divided by 9 leaves remainder 5 (since $9 \times 2 = 18$, $23 - 18 = 5$), so N also leaves 5. Then 23 divided by 3 leaves remainder 2 ($3 \times 7 = 21$, $23 - 21 = 2$), so N leaves 2. Taking the digit-sum of 23 again to get 5 and using that for the mod-3 answer is wrong because $5 \bmod 3$ is also 2 by luck but the reasoning differs; claiming the mod-3 remainder is 0 forgets 23 is not a multiple of 3; using remainder 2 for mod 9 confuses the two divisors.

2. (D) The digit-sum is $8 + A + 3 + 5 + B = 16 + A + B$, which must be a multiple of 9. With $B = A + 2$ the sum becomes $16 + A + (A + 2) = 18 + 2A$. For this to be a multiple of 9, $2A$ must be a multiple of 9; the only single-digit possibility keeping $B = A + 2$ a valid digit is $A = 0$ (sum 18, $B = 2$, giving 80352). $A = 9$ would need $B = 11$, impossible. So $A = 0$. The decoys 3, 1, 5 give digit-sums 24, 20, 28 — none divisible by 9, so each fails the divisibility-by-9 test.

3. (A) For a number with digits $d_3 d_2 d_1 d_0$, the value mod 11 equals the alternating sum $d_0 - d_1 + d_2 - d_3$. Reversing to $d_0 d_1 d_2 d_3$ gives alternating sum $d_3 - d_2 + d_1 - d_0$, the exact negative of the original. So M is congruent to $-7 \bmod 11$, and $-7 + 11 = 4$. Claiming reversal preserves the remainder is true only for palindromes or odd digit-counts; claiming the difference is always a multiple of 11 holds for the DIFFERENCE $N - M$, not for M itself; the remainder is fully determined as 4.

4. (C) An even product needs at least one even factor. An odd sum of three numbers needs an ODD count of odd numbers, so the number of odds is one or three. Three odds would make the product odd, contradicting the even product, so there is exactly one odd — and therefore exactly two even. Check: even + even + odd has one odd (odd count) giving an odd sum, and the two evens make the product even — valid. 'Exactly one even (two odds)' has two odds, an even count, giving an even sum — rejected. All even gives an even sum; all odd gives an odd product. Only exactly two even works.

5. (D) A number's digit-sum is a multiple of 3 exactly when the number itself is a multiple of 3 (the divisibility rule). From 1 to 60 the multiples of 3 are 3, 6, ..., 60, which number $60 / 3 = 20$. Answering 18 forgets to count both endpoints or miscounts the last term; 21 wrongly includes a non-multiple; 24 confuses the count with multiples of a smaller divisor. The digit-sum rule makes 'digit-sum divisible by 3' identical to 'number divisible by 3'.

6. (C) For a 3-digit number with hundreds digit a , tens b , units c , value $100a + 10b + c$; its reverse is $100c + 10b + a$. The difference is $99(a - c)$ in absolute value. Setting $99 |a - c| = 297$ gives $|a - c| = 3$. Answering 9 confuses the multiplier 99 with the digit gap; 33 divides 297 by 9 instead of 99; 2 misreads the factor. The tens digit cancels entirely.

7. (D) An odd product forces every one of the seven factors to be odd (a single even factor would make the product even). Seven odd numbers form an odd COUNT of odds, so their sum is odd; this holds regardless of the specific odd values. Saying it depends on values ignores that all seven are forced odd; restricting to all-ones is unnecessary since any seven odds work; an even sum is impossible because 7 is odd.

8. (B) Write the number with tens digit b and units c . Divisible by 9: $5 + 7 + b + c = 12 + b + c$ is a multiple of 9, so $b + c = 6$ or 15. Divisible by 11: the alternating sum (units + hundreds) - (tens + thousands) = $(c + 7) - (b + 5) = c - b + 2$ must be a multiple of 11, so $c - b = -2$ (taking 0) or 9. Combine $b + c = 6$ with $c - b = -2$: add to get $2c = 4$, $c = 2$, $b = 4$, giving 5742. Check: $5+7+4+2 = 18$ ($\div 9$) and $(2+7)-(4+5) = 0$ ($\div 11$). The branch $b + c = 15$ with $c - b = -2$ gives non-integers, and $c - b = 9$ forces $c > 9$. So 5742 is the unique answer. 5715 fails the 11-test, 5760 is not divisible by 11, and 5709 fails divisibility by 9.

9. (A) The multiples of 9 in this range run from 108 ($= 9 \times 12$) to 198 ($= 9 \times 22$). That is $22 - 12 + 1 = 11$ numbers. Answering 12 forgets that $100/9$ rounds up to 12 but $9 \times 11 = 99$ is below 100; 10 drops an endpoint; 9 miscounts the span. Listing 108, 117, ..., 198 confirms 11 terms.

10. (C) Let the number be $10t + u = 6(t + u)$, so $10t + u = 6t + 6u$, giving $4t = 5u$, i.e. $t/u = 5/4$. The single-digit solution is $t = 5$, $u = 4$, the number 54. Check: $6 \times (5 + 4) = 54$. It is even. The decoys 24, 42, 18 do not satisfy $10t + u = 6(t + u)$: $6 \times 6 = 36$ not 24; $6 \times 6 = 36$ not 42; $6 \times 9 = 54$ not 18. Only 54 fits both conditions.

11. (C) A six-digit repdigit equals $d \times 111111$, and $111111 = 111 \times 1001 = 111 \times 7 \times 11 \times 13$, so 111111 is divisible by 7. Therefore $d \times 111111$ is divisible by 7 for every digit d . Restricting to $d = 7$ confuses the digit with the divisor; requiring even digits invents an irrelevant parity condition; claiming none work ignores the factorisation $1001 = 7 \times 11 \times 13$.

12. (B) Let the first two terms be a and b . Then $\text{term}_3 = a + b = 11$, $\text{term}_4 = a + 2b$, $\text{term}_5 = 2a + 3b$, $\text{term}_6 = 3a + 5b = 47$. From $a + b = 11$ we get $a = 11 - b$; substitute: $3(11 - b) + 5b = 47$, so $33 + 2b = 47$, $b = 7$, and $a = 4$. Check term_6 : $3(4) + 5(7) = 12 + 35 = 47$, and $a + b = 11$ — both hold. So the 1st term is 4. A first term of 3 would give $\text{term}_6 = 49$; 5 gives 45; 2 gives 51.

13. (B) Because the units digits cycle with period 60, term 67 has the same units digit as term $67 - 60 = 7$. The 7th term is 13, units digit 3, so the 67th term also ends in 3. Answering 8 takes the 6th term; 1 takes the wrong offset; 6 is unrelated. Reducing the index modulo the period is the key step.

14. (A) In each block of 3 consecutive terms exactly one is even (the pattern is Odd, Odd, Even). $100 = 3 \times 33 + 1$, so there are 33 complete blocks contributing 33 evens, and the leftover 1 term is the first of the next block, which is Odd. Hence 33 evens. Answering 34 wrongly counts the leftover as even; 50 treats parity as alternating; the count is fixed by the period-3 pattern, not the start, so 'depends on start' is false for the standard 1,1,2 sequence.

15. (B) Work backwards using $\text{prev} = \text{current} - \text{previous-of-current}$. Before 72 (with 117 after it) the earlier term is $117 - 72 = 45$, so the run going back is ..., 45, 72, 117. Step back: before 45 is $72 - 45 = 27$; before 27 is $45 - 27 = 18$; before 18 is $27 - 18 = 9$. Counting four places before 72: that gives 45 (1 before), 27 (2 before), 18 (3 before), 9 (4 before). So the answer is 9. Answering 18 stops one step short; 27 stops two short; 12 mis-subtracts.

16. (A) The sum of the first n terms equals $\text{term}(n+2) - 1$. Here $n = 12$ terms are listed (ending at $144 = \text{term}_{12}$), so the sum = $\text{term}_{14} - 1$. Extend the sequence: $\text{term}_{13} = 89 + 144 = 233$, $\text{term}_{14} = 144 + 233 = 377$. Thus the sum = $377 - 1 = 376$. The value 375 is one short, 232 mistakes it for $\text{term}_{13} - 1$, and 377 forgets to subtract 1.

17. (B) The terms are $a, b, a+b, a+2b, 2a+3b$, so the 5th term gives $2a + 3b = 30$. Then $2a = 30 - 3b$ must be positive and even, so b is even and $b \leq 9$: b in $\{2, 4, 6, 8\}$. $b = 2$ gives $a = 12$ ($a > b$, rejected); $b = 4$ gives $a = 9$ ($a > b$, rejected); $b = 6$ gives $a = 6$ ($a = b$, valid); $b = 8$ gives $a = 3$ ($a < b$, valid). So the valid pairs are (6,6) and (3,8) — exactly 2. Answering 3 or 4 counts the rejected $a > b$ pairs; 1 misses one valid pair.

18. (D) Tilings of length n follow the rule $t(n) = t(n-1) + t(n-2)$: $t(7) = 21$, $t(8) = 34$, $t(9) = 55$, $t(10) = 89$. Exactly one tiling uses no 2-tile at all (all ten 1-tiles). So tilings with at least one 2-tile = $89 - 1 = 88$. Answering 89 forgets to remove the all-singles tiling; 55 gives $t(9)$; 87 subtracts two by mistake. Only one all-1-tile arrangement exists, hence subtract exactly one.

19. (D) Going from the 9th to the 15th term is 6 steps, so the growth factor is about 1.618 to the 6th power. 1.618 squared is about 2.618, cubed about 4.24, to the 4th about 6.85, 5th about 11.1, 6th about 17.9 — close to 18. Answering 6 times counts only 4 steps; 1.6 times counts a single step; 11 times uses 5 steps instead of 6. Each step multiplies by roughly 1.618, and there are six of them.

20. (C) If two consecutive terms are both multiples of 3, their sum is also a multiple of 3, so by induction every term stays a multiple of 3 once the first two (both 3) are. The sequence 3,3,6,9,15,24,39,... is entirely divisible by 3. Picking the 4th, 6th or 7th term assumes the multiple-of-3 property eventually breaks, but addition preserves a common factor of the seeds forever.

21. (C) By the sum identity, the sum of the first 17 terms = term₁₉ - 1. Parity of term₁₉: the pattern O,O,E has period 3, and $19 = 3 \times 6 + 1$, so term₁₉ is the first in a block, which is Odd. Then term₁₉ - 1 is even. So the sum is even. Answering odd forgets the -1 flips parity; the parity is fully determined by the period-3 pattern, so 'cannot be determined' is wrong.

22. (A) The next term is $105 + 170 = 275$. Its units digit 5 makes it odd. Answering 275 even ignores the actual digit; 265 mis-adds; 65 subtracts instead of adds ($170 - 105$). The forward rule adds the two previous terms, giving 275, an odd number.

23. (B) The sum of all calls counts exactly the 'taller-ahead' pairs (each call at a position tallies how many earlier children exceed that child's height). The sum $0 + 1 + 2 + 3 + 4 + 5 = 15$. This is the maximum, occurring when heights strictly decrease front to back. Answering 21 uses $6 \times 7 / 2$; 5 takes only the last call; 10 halves the count. The total of the calls IS the inversion count, namely 15.

24. (A) Calls 0,1,2,3,4 mean each child has more taller children ahead than the one before, so heights strictly decrease from front to back; the front child is tallest and the back child is shortest. Hence the shortest stands 5th (back). Saying front confuses tallest with shortest; middle ignores the strict ordering; the configuration is forced, so it is determinable.

25. (C) A call of $k - 1$ at the last position means every one of the $k - 1$ children ahead is taller, so the back child is shorter than all of them — the shortest in the line. Calling it tallest reverses the meaning; median is unsupported; the conclusion is certain because $k - 1$ is the maximum possible call and saturates it.

26. (A) Every call is 0, meaning no child has a taller child ahead, so heights strictly INCREASE front to back and there are no taller-ahead pairs at all. The sum of calls $0+0+\dots+0 = 0$ equals the pair count. Answering 7 or 6 invents pairs; 21 gives the maximum for the reversed order. All-zero calls mean a fully sorted-ascending line, zero inversions.

27. (C) The bound at position k is $k - 1$: position 1 max 0, position 2 max 1, position 3 max 2, position 4 max 3, position 5 max 4. The calls 0,1,1,2,0 satisfy $0 \leq 0, 1 \leq 1, 1 \leq 2, 2 \leq 3, 0 \leq 4$ — all within bounds. So none is impossible. Flagging the 2nd, 4th or 5th child mistakes a valid call for a violation; the sequence is realisable.

28. (A) The sum of all calls equals the number of taller-ahead pairs. Sum = $0 + 1 + 2 + 0 + 1 + 2 + 3 = 9$. Answering 6 drops some terms; 12 over-adds; 21 gives the maximum for a fully reversed line of 7. Simply totalling the calls yields the inversion count, 9.

29. (C) Calls 0,1,...,8 for $n = 9$ mean heights strictly decrease front to back. The middle position is the 5th, whose call is 4 — meaning 4 taller children stand ahead. Since heights strictly decrease, no child behind the 5th is taller, so exactly 4 children in total are taller than the middle child. Answering 5 off-by-ones the position; 8 gives the back child's count; 0 gives the front child's. The call at the middle position directly states the answer, 4.

30. (C) Process front to back assigning ranks (1 = shortest among placed so far). Call 0 for child1: no taller ahead. Child2 call 1: one taller ahead (child1 taller than child2). Child3 call 0: no taller ahead, so child3 is taller than both child1 and child2 placed before it. Child4 call 2: exactly two of the three ahead are taller. Since child3 is tallest so far and $\text{child1} > \text{child2}$, the two taller-than-child4 are child3 and child1, so child4 sits below child1 but above child2; final order tallest-to-shortest: $\text{child3} > \text{child1} > \text{child4} > \text{child2}$. The tallest is child3, at the 3rd position. Answering 1st, 4th or 2nd mis-tracks the running comparisons.

31. (B) The sum of calls is the inversion count, maximised when every earlier child is taller than every later one — heights strictly decreasing. Then position k contributes $k - 1$, totalling $0 + 1 + \dots + (n - 1) = n(n-1)/2$. The formula $n(n+1)/2$ over-counts by including a phantom pair; $n - 1$ is just the single largest call; n squared exceeds the maximum possible pair count. The decreasing arrangement saturates every pair as an inversion.

32. (D) For 6 children the call-sum (inversion count) ranges from 0 (sorted ascending) to $6 \times 5 / 2 = 15$ (sorted descending), and every integer in between is achievable by a suitable permutation. 14 is in $[0, 15]$, so it is possible. The total need not be a multiple of 6; the maximum is 15 not 10; distinct heights are required by the game, so 'sharing a height' is irrelevant and false.

33. (C) Corners have 2 edge-neighbours, edge-middles have 3, the centre has 4 — fewest is a corner with 2. But a supercell must EXCEED all neighbours, and 1 is the smallest of 1-9, so it can never exceed any neighbour, regardless of position. The centre has the MOST

neighbours, not fewest; an edge-middle has 3; and no placement makes 1 a supercell. Fewest neighbours = corner (2), and 1 is never a supercell.

34. (D) In the standard 3x3 magic square the centre is always $15 / 3 = 5$ (the average of 1-9), and the even numbers 2,4,6,8 sit at the edge-middles while odd 1,3,7,9 sit at the corners. So the centre is 5. Answering 1 or 9 violates the centre-equals-5 theorem (any line through the centre forces it to the mean); the centre is fixed, so 'it varies' is false.

35. (A) The whole grid uses 1-9, total 45, so the three row sums add to 45. Given 7 and 23, the middle row is $45 - 7 - 23 = 15$. A top row summing to 7 (e.g. 1+2+4) and a bottom summing to 23 (e.g. 9+8+6) leave middle digits 3,5,7 summing to 15 — possible. Answering 14 or 16 mis-subtracts; claiming 15 impossible ignores that the leftover three digits genuinely sum to 15.

36. (B) Corners have the fewest neighbours (2 each), so placing the largest values there lets four cells each exceed both their neighbours simultaneously, giving the maximum of 4 supercells (e.g. 9,7,8,6 in corners with smaller numbers on edges/centre). Edge-middles have 3 neighbours and the centre has 4, harder to dominate, and adjacent supercells in one row would conflict (neighbours). The four corners is the optimal configuration.

37. (D) Every rotation and every reflection of a magic square is again a magic square (the eight such symmetries are exactly the dihedral group of the square); each one sends the set of lines summing to 15 to itself. Doing two of them in a row still yields a magic square. The diagonals are preserved (they map to diagonals or to each other); the centre stays central under any symmetry; the magic sum 15 is unchanged regardless of its parity.

38. (B) Each diagonal is (corner) + 5 + (opposite corner). Diagonal one = 15 means its two corners sum to $15 - 5 = 10$; diagonal two = 19 means its two corners sum to $19 - 5 = 14$. The four corners are exactly these two diagonal pairs, summing to $10 + 14 = 24$. Answering 20 forgets one diagonal's excess; 26 double-counts the centre; 21 mis-adds. Subtract the shared centre once per diagonal, then add.

39. (D) The four largest digits are 6,7,8,9 summing to 30, the maximum any four cells (hence the corners) can reach. 28 is below 30 and achievable, e.g. corners 9,8,7,4 or 9,8,6,5. Claiming the max is 24, 26 or 28 understates it; the four-largest bound is $6+7+8+9 = 30$, so 28 fits.

40. (B) The three row sums total 45 (sum of 1-9). If all three could be 15 the largest would be 15; the standard magic square 8-1-6 / 3-5-7 / 4-9-2 achieves all rows = 15, so the largest row is 15. You cannot do better: three rows summing to 45 must have a largest of at least $45 / 3 = 15$. Answering 14 is below the forced average; 16 is achievable but not minimal; 24 is the maximum row, not the minimised largest.

41. (B) The left column = (top-left) + (centre) + (bottom-left) = $4 + 5 + (\text{bottom-left}) = 18$, so bottom-left = 9. The remaining two left-column cells are the centre 5 and the bottom-

left 9, summing to $5 + 9 = 14$. Answering 11 forgets the corner is excluded from 'remaining'; 13 mis-adds; 9 gives only the bottom-left. Subtract the corner from the column total to confirm, then add the two non-corner cells.

42. (A) 9 is the largest, so it is a supercell wherever it goes; 8 is a supercell only if 9 is not one of its neighbours (since $9 > 8$). So 9 and 8 must not share an edge — two opposite corners work, each dominating its own two neighbours. Putting 9 in the centre makes it adjacent to potentially 8 and bars 8 nearby; placing them side by side makes 8 lose to its neighbour 9; centre-8 is dominated by an adjacent larger cell. Non-adjacent placement is required.

43. (D) Replacing each value v by $16 - v$ sends 1-9 to 15,14,...,7 — still nine distinct values, and each line of three now sums to $3 \times 16 - 15 = 48 - 15 = 33$. Since every original line summed to 15, every new line sums to 33, so it remains a magic square (this is the complement map). The sum is not unchanged (each cell changed); the diagonals survive because the transformation is linear and uniform; 48 forgets to subtract the old sum.

44. (D) From 7: 7->22->11->34->17->52->26->13->40->20->10->5->16->8->4->2->1. Counting the arrows: 7->22 (1), ->11 (2), ->34 (3), ->17 (4), ->52 (5), ->26 (6), ->13 (7), ->40 (8), ->20 (9), ->10 (10), ->5 (11), ->16 (12), ->8 (13), ->4 (14), ->2 (15), ->1 (16). So 16 steps. Answering 17 over-counts; 11 stops at 5; 15 stops at 2.

45. (D) The Collatz trajectory from 27 is famously long (111 steps) and peaks at 9232 before descending to 1. This is a well-known landmark value. 4616 is half of the peak (one halving past it); 6174 is the unrelated Kaprekar constant; 3077 is not on the trajectory. The maximum reached is 9232.

46. (A) Step 1 on 3524: descending 5432, ascending 2345, difference $5432 - 2345 = 3087$. Step 2 on 3087: descending 8730, ascending 0378, difference $8730 - 378 = 8352$. So after two steps the value is 8352. (A third step, $8532 - 2358$, would give the famous 6174.) Thus 3087 is the one-step result, 6174 the three-step fixed point, and 2475 is unrelated.

47. (B) From 23: $2^2 + 3^2 = 4 + 9 = 13$ (step 1); $1^2 + 3^2 = 1 + 9 = 10$ (step 2); $1^2 + 0^2 = 1$ (step 3). So 23 reaches 1 in exactly 3 steps — it is a 'happy number'. Numbers that are NOT happy fall into the cycle 4,16,37,58,89,145,42,20,4..., but 23 escapes to 1, so the 'never reaches 1' option is wrong here; 5 and 8 steps over-count.

48. (C) Trace: 50 (even) ->25 (step1); 25 (odd) ->22 (step2); 22 (even) ->11 (step3); 11 (odd) ->8 (step4); 8 (even) ->4 (step5); 4 (even) ->2 (step6). After exactly 6 steps the value is 2. Answering 1 over-runs by a step; 4 stops at step 5; 0 is never reached.

49. (A) 12->6->3->10->5->16->8->4->2->1. Steps: 12->6 (halve), 6->3 (halve), 3->10 ($3n+1$, odd step), 10->5 (halve), 5->16 ($3n+1$, odd step), 16->8 (halve), 8->4 (halve), 4->2 (halve), 2->1 (halve). That is 9 steps total; the two odd ($3n+1$) steps are 3->10 and 5->16,

so $9 - 2 = 7$ halving steps. Answering all 9 halving forgets the two tripling steps; 10 steps over-counts; 8 steps stops early.

50. (D) Step 1 on 297: descending 972, ascending 279, $972 - 279 = 693$. Step 2 on 693: $963 - 369 = 594$. Step 3 on 594: $954 - 459 = 495$, the fixed point. So it takes 3 steps. Stopping at 693 (2 steps) or 594 (which would be the 2-step value) under-counts; 4 over-counts since 495 is reached and stays put at step 3; 1 step gives only 693.

51. (D) The digital root equals the number mod 9 (using 9 for multiples of 9). 4004's digit-sum is $4 + 0 + 0 + 4 = 8$, already a single digit, so the digital root is 8. Equivalently $4004 \bmod 9$: $4004 = 9 \times 444 + 8$. Answering 9 would require a multiple of 9; 2 and 4 mis-add the digits. The digital root is 8.

52. (A) A palindrome abba has alternating digit-sum $a - b + b - a = 0$, which is a multiple of 11, so EVERY 4-digit palindrome is divisible by 11. The 4-digit palindromes have a from 1-9 (9 choices) and b from 0-9 (10 choices), giving $9 \times 10 = 90$. Answering 9 counts only one b; 18 counts two; 45 halves wrongly. All 90 are divisible by 11.

53. (C) A palindrome abba is divisible by 9 when $2a + 2b$ is a multiple of 9, i.e. $a + b$ is a multiple of 9 (since $2(a+b)$ multiple of 9 needs $a+b$ multiple of 9 as $\gcd(2,9)=1$). Divisible by 4 needs the last two digits 'ba' ($10b + a$) divisible by 4. We want the largest, so maximise a. $a + b$ multiple of 9 with a as large as possible: $a = 9$ needs $b = 0$ or 9. $a=9, b=0 \rightarrow 9009$; last two digits 09 $\rightarrow 9$ not divisible by 4. $a=9, b=9 \rightarrow 9999$; last two 99 not divisible by 4. $a=8$ needs $b=1$ (sum 9) or $b=10$ (no): 8118; last two 18 $\rightarrow$ not divisible by 4. $a=7, b=2 \rightarrow 7227$, last two 27 no. $a=6, b=3 \rightarrow 6336$, last two 36 $\rightarrow 36/4 = 9$, divisible! Check 9: $6+3+3+6 = 18$, divisible by 9. So 6336 works. Larger candidates (9009, 8118, 7227) all fail the divisible-by-4 test, and 9999/8888 are not divisible by 9 and 4 respectively. The largest qualifying palindrome is 6336.

54. (A) A two-digit number $10t + u$ plus its reverse $10u + t$ equals $11(t + u)$. Setting $11(t + u) = 121$ gives $t + u = 11$. The digits differ by 3, so combining $t + u = 11$ with $t - u = 3$ gives $t = 7$, $u = 4$. The number is 74 (or its reverse 47); the larger is 74, and $74 + 47 = 121$ with digit gap 3. The decoy 85 gives $85 + 58 = 143$; 63 gives $63 + 36 = 99$; and a valid pair clearly exists, so 'no such number' is false.

55. (A) Largest arrangement 8642, smallest 2468; difference $8642 - 2468 = 6174$. Its digit-sum $6 + 1 + 7 + 4 = 18$, a multiple of 9, so 6174 is divisible by 9 (indeed any number minus its digit-rearrangement is divisible by 9 since rearranging preserves the digit-sum). 6084 and 5994 are arithmetic slips; the difference IS divisible by 9, so the 'not divisible' option is false. The answer is 6174, divisible by 9.

56. (C) Divisibility by 4 depends on the last two digits, which for aba are 'ba' $= 10b + a$. We need $10b + a$ divisible by 4 for a in 1-9, b in 0-9. For each fixed a, $10b + a \bmod 4 = (2b + a) \bmod 4$ (since $10 = 8 + 2$, $10b \equiv 2b \bmod 4$). We need $2b + a \equiv 0 \bmod 4$. For a even ($a =$

2,4,6,8): $a \bmod 4$ is 2,0,2,0; we need $2b \equiv -a \bmod 4$. $2b \bmod 4$ is 0 (b even) or 2 (b odd). $a=2$ needs $2b \equiv 2 \rightarrow b$ odd: 5 values; $a=4$ needs $2b \equiv 0 \rightarrow b$ even: 5 values; $a=6$ needs $2b \equiv 2 \rightarrow b$ odd: 5; $a=8$ needs $2b \equiv 0 \rightarrow b$ even: 5. Odd a (1,3,5,7,9): a is odd so $2b + a$ is odd, never divisible by 4: 0 values. Total = $5 \times 4 = 20$. Answering 18 or 25 miscounts the b-parity cases; 10 counts only half the even- a values. The answer is 20.

57. (D) The original minus its reverse is $99(a - c)$ where $a > c$, giving one of 99,198,...,891 — all 3-digit (after the middle becomes 9) of the form where the result is X. The classic result: original - reverse always yields a number whose reverse added back gives 1089 (e.g. $853 - 358 = 495$; $495 + 594 = 1089$). This is the famous '1089 trick'. Answering 99 or 198 stops at the subtraction stage; 1000 is a near-miss. Adding the reversal of the difference always lands on 1089.

58. (B) A 4-digit palindrome in (1000, 2000) has form 1bb1 (first and last digit 1, since the number is between 1000 and 2000), with the middle two digits equal: 1001, 1111, 1221, ..., 1991. The middle digit b runs 0-9, giving 10 palindromes. But 1001 is greater than 1000 (strictly between holds) and $1991 < 2000$. So 10 of them. Answering 9 drops an endpoint value; 11 over-counts; 100 ignores the palindrome constraint. There are 10.

59. (B) abcba is divisible by 5 when its last digit a is 0 or 5. But a is also the first digit, which cannot be 0 (or it would not be a 5-digit number), so a must be 5. With $a = 5$ fixed, b ranges over 0-9 (10 choices) and c over 0-9 (10 choices), giving $10 \times 10 = 100$ palindromes. Answering 200 wrongly allows $a = 0$ as well; 180 mixes the counts; 90 drops one value of b .

60. (B) There are $5! = 120$ arrangements of five distinct digits. A number's parity depends only on its units digit; since every digit 1,3,5,7,9 is odd, the units digit is always odd, so every one of the 120 numbers is odd. Answering 60 assumes half are even (impossible with only odd digits); 24 fixes one position needlessly; 'none guaranteed' contradicts that all digits are odd. All 120 are odd.